Stefan Milenković

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Education

EPFL — Lausanne, Switzerland
PhD in Computer Science, EDIC Fellowship — *September 2026 – present*

Faculty of Mathematics, University of Belgrade — Belgrade, Serbia
MSc in Mathematics and Computer Science, GPA: 9.5/10.0 — *2024 – 2026*

Faculty of Mathematics, University of Belgrade — Belgrade, Serbia
BSc in Mathematics and Computer Science, GPA: 10.0/10.0 — *2020 – 2024*

Research Experience

Research Intern (Summer@EPFL) — June 2025 – August 2025
École Polytechnique Fédérale de Lausanne — *Lausanne, Switzerland*

- Proved properties of the non-classical logic Grounded Arithmetic using Isabelle/HOL
- Worked on proving consistency of the logic and primitive-recursiveness of relevant functions
- Supervised by Prof. Bryan Alexander Ford

Compiler Researcher — March 2025 – May 2026
Oracle Labs — *Belgrade, Serbia*

- Worked in the GraalVM Native Image compiler research group
- Improved binary size reduction in the GraalVM Native Image compiler by incorporating method names into existing ML models for call count prediction
- Evaluated the performance impact of various optimizations in the GraalVM Native Image compiler

Summer Research Internship — 2021
Mathematical Institute of the Serbian Academy of Sciences and Arts (MISANU) — *Belgrade, Serbia*

- Designed and implemented a theorem prover for propositional logic based on the method of analytic tableaux

Publications & Talks

Dimitrios Alexopoulos, Elliot Bobrow, Yusuf Demir, Bryan Ford, **Stefan Milenković**, Omar Muhammad, Ananthajit Srikanth, *Have a thing? Reasoning around recursion with dynamic typing in grounded arithmetic*. *arXiv preprint* arXiv:2510.25369, 2025

Stefan Milenković, Milan Čugurović, Milena Vujošević Janičić, *Improving ML-Based Static Profiling Using Method Names*, XV Symposium "Mathematics and Applications", Belgrade, Serbia. Talk, December 2025

Stefan Milenković, Antonije Subotić, *Method of Analytic Tableaux in Propositional Logic*, Student Seminar, MISANU, Belgrade, Serbia. Talk, December 2021 (in Serbian)

Work Experience

Teaching Assistant — November 2024 – August 2026
Faculty of Mathematics, University of Belgrade — *Belgrade, Serbia*

- Delivered exercise sessions for the courses *Programming Language Translation* and *Lexical Analysis*
- Wrote an online supplementary course material for Abstract Syntax Tree construction (in Serbian)

Projects

Semantic Checks within Clang | *LLVM, C++* — 2025

- Improved the existing Clang checker for detecting unsafe functions by making the list of functions extendable

LLVM Inline Pass | *LLVM, C++* — 2024

- Developed an LLVM compiler pass to inline function calls, optimizing runtime performance

Achievements

Best Computer Science Graduate, Class of 2024 — 2025

- Annual faculty award *Nedeljko Parezanović* awarded to the best Computer Science graduate of the year

Languages

English (C1, IELTS: 8.0/9.0), Serbian (native)